Juan Diego Ron Molina

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Fourth-year Computer Engineering student at Gazi University with hands-on experience in machine learning, computer vision, agentic LLM systems, and deploying AI models in production.

SKILLS

- **Programming:** *Python, R, Java, C/C++, C#, SQL, Git/GitHub.*
- **Machine Learning & Data Science:** *scikit-learn, PyTorch, TensorFlow, Pandas, NumPy, Data Visualization, NLP, Computer Vision, OpenCV, MediaPipe, Ultralytics YOLO, Synthetic Data Generation, Model Optimization.*
- **Frameworks & Tools:** *HuggingFace, TensorRT, ONNX, GStreamer, RTSP, BlenderProc, SpringBoot, Flask, REST APIs, Web Scraping, Matplotlib, HTML5, CSS3, Version Control.*
- **LLMs & Agents:** *LangGraph, LangChain, Ollama, Langfuse, Pydantic.*
- **Data & DevOps:** *PostgreSQL, pgvector, SQLAlchemy, Alembic, Docker Compose, GitHub Actions, pytest, ruff, mypy, uv.*

EXPERIENCE

- **YourEye Technologies** — Artificial Intelligence Engineer. ***May 2026 – Present.***
  - Engineered TensorRT workflows converting PyTorch/Ultralytics YOLO models (.pt) into FP16 TensorRT engines via ONNX for live camera analytics GPU servers, reducing inference latency and GPU memory load.
  - Deployed computer vision models on NVIDIA RTX PRO 6000 Blackwell GPU servers by building RTSP GStreamer pipelines that stream live security-camera feeds into real-time inference.
  - Developed a computer vision system recognizing real parts using only BlenderProc synthetic data from client CAD models; a detector + DINOv2 matcher enrolls new CADs in minutes without retraining and rejects unknown parts.
  - Identified 25/25 parts under unseen viewpoints, finishes, and lighting, and 92/100 on complex backgrounds (0 wrong labels); cut inference from ~620 to ~145 ms/frame (laptop → server GPU) for live phone-camera use.
  - Built a real-time face recognition system using InsightFace (buffalo_l), combining a RetinaFace detector with an ArcFace ResNet-50 embedder to produce 512-dimensional L2-normalized face embeddings.
- **Syntexchub** — Artificial Intelligence Intern. ***Mar 2026 – Apr 2026.***
  - Built a hand gesture recognition system using OpenCV and MediaPipe, processing real-time webcam input to classify distinct hand gestures.
  - Designed a voice-enabled chatbot using Llama (HuggingFace) for responses and OpenAI's Whisper-large-v3-turbo for speech-to-text, with prompt engineering and context handling for multi-turn conversations.
  - Implemented a chain-of-thought prompting pipeline that broke complex queries into intermediate reasoning steps, improving response quality for Explainable AI use cases.

EDUCATION

- **B.E. Computer Engineering** — Gazi University, Ankara. (G.P.A: 3.41/4.00.) ***Sep 2023 – Jul 2027.***

SCHOLARSHIPS

- **Turkish Scholarships Recipient** — Bachelor's degree, fully funded. ***Aug 2022 – Jul 2027.***

PROJECTS

- **HomeOps** — Multi-agent household operations assistant, used through Telegram, running fully on local open-weight LLMs (inventory agent and receipt pipeline built; meal-planning, procurement, and finance agents planned). ***Oct 2026 – Present.***
  - Built a LangGraph inventory agent on a local qwen3:8b model in which the LLM picks tools and deterministic Python tools make every pantry change, with state saved in Postgres (PostgresSaver); reached a 97% pass rate (35/36) on a Spanish chat benchmark at 1.4 s median per message.
  - Developed a receipt-to-pantry pipeline for Mercadona, Lidl, and Dia receipts where qwen2.5vl:7b extracts JSON-schema-constrained line items and Python code verifies the numbers; on 12 synthetic receipts (120 lines), found 100% of lines with none invented and 92% of quantities and prices correct.
  - Built a Telegram bot (python-telegram-bot) with a user allow-list, confirm/cancel buttons, /despensa and /caduca commands, and free-text Spanish corrections, saving nothing until the user confirms.
  - Designed expiry estimation from a curated shelf-life table (~45 categories) in which the LLM may only pick a schema-constrained category, so it cannot invent a shelf life.
  - Ran 100% open-weight models locally through Ollama's OpenAI-compatible API on an 8 GB RTX 4060, running text and vision models one at a time (~9 s swap); scored 20/20 on a tool-calling benchmark at ~46 tokens/s.
  - Applied eval-driven development, with every agent shipping a benchmark and a ≥ 90% pass threshold, plus Langfuse tracing, Docker Compose, Alembic migrations, and GitHub Actions CI (ruff, mypy, pytest).
- **Unmanned Ground Vehicle** — Software for an autonomous ground vehicle's navigation and control, with planned participation in TEKNOFEST. ***Dec 2025 – Sep 2026.***
  - Developed the software interface for vehicle monitoring and control, enabling real-time visualization of navigation data and system status.
  - Performed data annotation on image datasets to support computer vision model training for object detection and environmental perception tasks.

- **Crime Rate Factors** — Co-authored research paper analyzing socio-economic predictors of U.S. violent crime using the UCI *Communities and Crime* dataset (1,994 samples, 128 features). ***Apr 2025 – May 2025.***
  - Designed an end-to-end ML pipeline: data cleaning, Random Forest imputation, outlier removal, feature selection, and SMOTE for class imbalance, reducing the dataset to 68 informative features.
  - Benchmarked 15 regression and classification algorithms (incl. Random Forest, SVR, CatBoost, ANN) using ROC-AUC and 5-fold cross-validation; selected an ANN as the final model (96.33% accuracy), prioritizing generalization over higher-variance boosting models.
- **MiKuri** — Pokémon TCG-inspired card battle game built in Java for an OOP course. ***Nov 2024 – Dec 2024.***
  - Designed the game's class hierarchy using OOP principles (inheritance, polymorphism, encapsulation) to model cards, players, decks, and battle mechanics.
  - Implemented the user interface with JavaFX and SceneBuilder, using FXML for layout and controller classes for game logic, following an MVC-style separation.

CERTIFICATES

- **Machine Learning A-Z**, SuperDataScience (*Sep 2025*) • **The Complete Python Pro Bootcamp**, London App Brewery (*Aug 2025*).
- **Google Cloud Skills Boost**, Google (*Aug 2025*) • **Introduction to Data Science**, Cisco (*Mar 2025*) • **Claude Code 101**, Anthropic (*Jun 2026*).

LANGUAGES

Spanish (native), **English** (bilingual), **Turkish** (professional working proficiency), **French** (basic proficiency).